I- Periodic functions

Periodic phenomena occur all the time in the physical world. For example, in :

— variations in average maximum and minimum monthly temperatures

— tidal variations in the depth of water in a harbour

— the phases of the moon

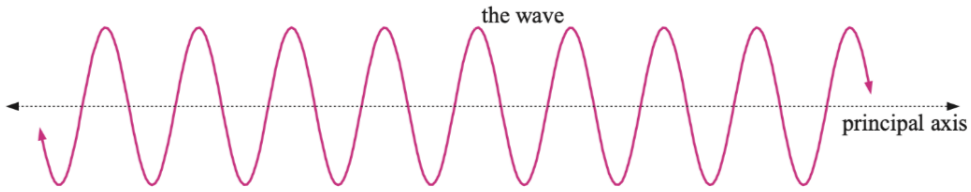
These phenomena illustrate variable behaviour which is repeated over time. The repetition may be called periodic, oscillatory, or cyclic in different situations.

a) Vocabulary

Definition

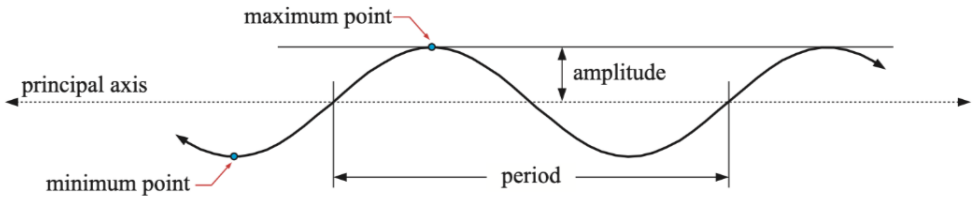
A periodic function is one which repeats itself over and over in a horizontal direction, in intervals of the same length.

Example



Definition

The period of a periodic function is the length of one repetition or cycle.



Definition

$f(x)$ is a periodic function with period $p \iff f(x+p) = f(x)$ for all x, and p is the smallest positive value for this to be true.

Definition

The wave oscillates about a horizontal line called the principal axis or mean line which has equation $y = \frac{\max + \min}{2}$.

Remark. *A maximum point occurs at the top of a crest, and a minimum point at the bottom of a trough.*

Definition

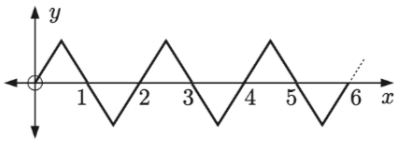
The amplitude is the distance between a maximum (or minimum) point and the principal axis.

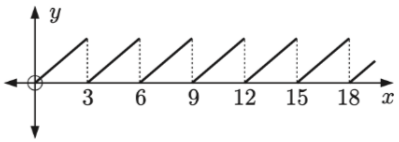
amplitude $= \frac{\max - \min}{2}$.

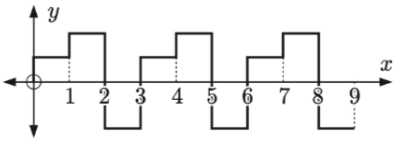
b) Exercises Set I

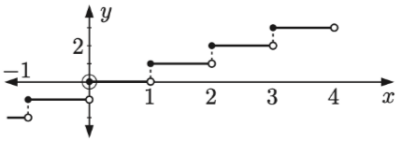
EXERCISE 1

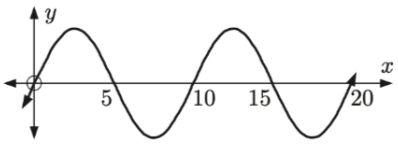
Which of these graphs show periodic behaviour?

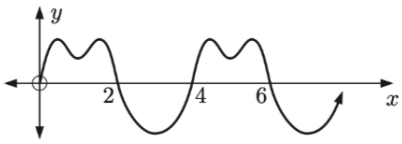
a 

b 

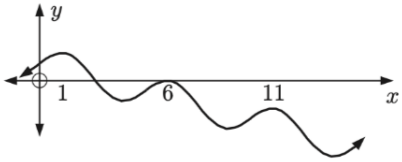
c 

d 

e 

f 

g 

h 

EXERCISE 2

The following tabled values show the height above the ground of a point on a bicycle wheel as the bicycle is wheeled along a flat surface.

Distance travelled (cm)	0	20	40	60	80	100	120	140	160	180	200
Height above ground (cm)	0	6	23	42	57	64	59	43	23	7	1

Distance travelled (cm)	220	240	260	280	300	320	340	360	380	400
Height above ground (cm)	5	27	40	55	63	60	44	24	9	3

a Plot the graph of height against distance.

b Is it reasonable to fit a curve to this data, or should we leave it as discrete points?

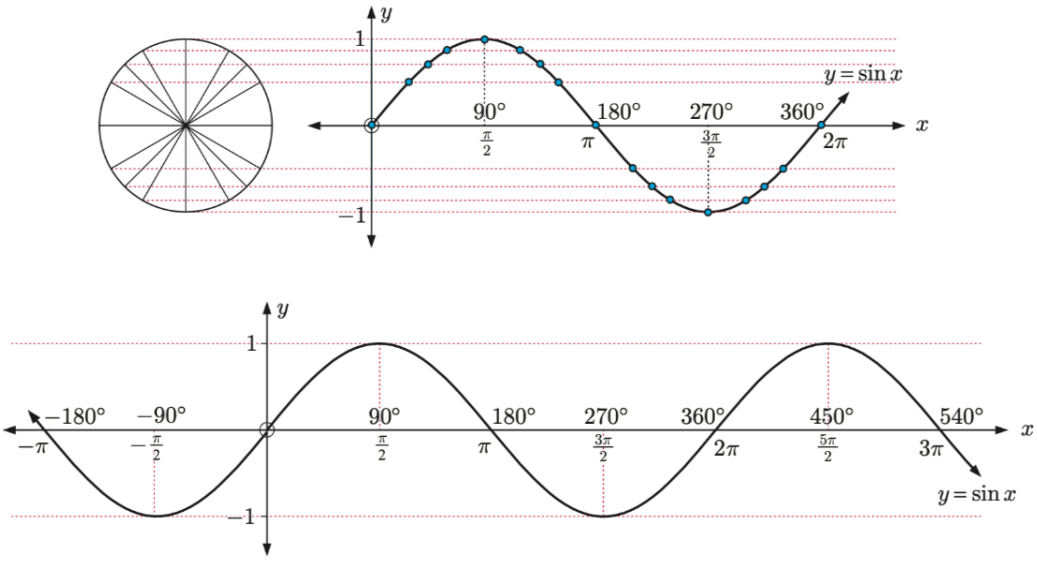
c Is the data periodic? If so, estimate:

i the equation of the principal axis ii the maximum value

iii the period iv the amplitude.

II- Sine functions

Basic sine curve



The unit circle repeats itself after one full revolution, so its *period* is 2π.

The *maximum* value is 1 and the *minimum* is -1, as $-1 \leqslant y \leqslant 1$ on the unit circle.

The *amplitude* is 1.

Investigation

How does a affect the function $y = a\sin x$?

State the amplitude of :

1. $y = 3\sin x$
2. $y = \sqrt{7}\sin x$
3. $y = -2\sin x$

How does b affect the function $y = \sin bx$?

State the period of :

1. $y = \sin 3x$
2. $y = \sin\left(\frac{1}{3}x\right)$
3. $y = \sin(1.2x)$
4. $y = \sin bx$

a) Transformation

Property

Family $y = a\sin x, \quad a \neq 0$

— a affects the amplitude of the graph ; amplitude $= |a|$

— The graph is a vertical stretch of $y = \sin x$ with scale factor $|a|$.

— If $a < 0$, the graph of $y = \sin x$ is also reflected in the x-axis.

Property

Family $y = \sin bx, \quad b > 0$

— The graph is a horizontal stretch of $y = \sin x$ with scale factor $\frac{1}{b}$.

— period $= \frac{2\pi}{b}$

Example

Without using technology, sketch the graphs of :

1. $y = 2\sin x$
2. $y = -2\sin x$ for $0 \leq x \leq 2\pi$.

Example

Without using technology, sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$.

Investigation

1. What transformation moves $y = \sin x$ to $y = \sin(x - c)$?
2. What transformation moves $y = \sin x$ to $y = \sin x + d$?
3. What transformation moves $y = \sin x$ to $y = \sin(x - c) + d$?

Property

— $y = \sin(x - c)$ is a horizontal translation of $y = \sin x$ through c units.

— $y = \sin x + d$ is a vertical translation of $y = \sin x$ through d units.

— $y = \sin(x - c) + d$ is a translation of $y = \sin x$ through vector $\begin{pmatrix} c \\ d \end{pmatrix}$.

Example

On the same set of axes graph for $0 \leq x \leq 4\pi$:

1. $y = \sin x$ and $y = \sin(x - 1)$
2. $y = \sin x$ and $y = \sin x - 1$

Property

The general sine function is $y = a\sin\big(b(x - c)\big) + d$ where $b > 0$.

a affects amplitude b affects period c affects horizontal translation d affects vertical translati

The principal axis of the general sine function is $y = d$.

The period of the general sine function is $\dfrac{2\pi}{b}$.

The amplitude of the general sine function is $|a|$.

When patterns of variation can be identified and quantified using a formula or equation, predictions may be made about behaviour in the future. Examples of this include tidal movement which can be predicted many months ahead, and the date of a future full moon.

b) Exercises Set II

EXERCISE 3

Without using technology, sketch the graphs of the following for $0 \leqslant x \leqslant 2\pi$:

a $y = 3\sin x$ b $y = -3\sin x$ c $y = \frac{3}{2}\sin x$ d $y = -\frac{3}{2}\sin x$

EXERCISE 4

State the period of:

a $y = \sin 4x$ b $y = \sin(-4x)$ c $y = \sin\left(\frac{x}{3}\right)$ d $y = \sin(0.6x)$

EXERCISE 5

Find b given that the function $y = \sin bx$, $b > 0$ has period:

a 5π b $\frac{2\pi}{3}$ c 12π d 4 e 100

EXERCISE 6

State the period of:

a $y = \sin 5t$ b $y = \sin\left(\frac{t}{4}\right)$ c $y = \sin(-2t)$

EXERCISE 7

Find b in $y = \sin bx$ if $b > 0$ and the period is:

a 3π b $\frac{\pi}{10}$ c 100π d 50

EXERCISE 8

State the transformation(s) which map:

a $y = \sin x$ onto $y = \sin x - 1$

b $y = \sin x$ onto $y = \sin(x - \frac{\pi}{4})$

c $y = \sin x$ onto $y = 2\sin x$

d $y = \sin x$ onto $y = \sin 4x$

e $y = \sin x$ onto $y = \frac{1}{2}\sin x$

f $y = \sin x$ onto $y = \sin\left(\frac{x}{4}\right)$

g $y = \sin x$ onto $y = -\sin x$

h $y = \sin x$ onto $y = -3 + \sin(x + 2)$

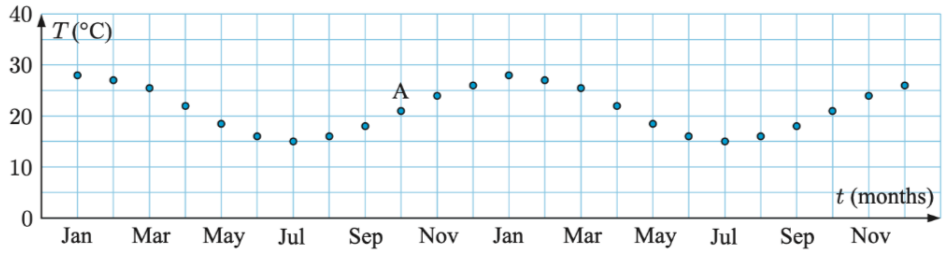
i $y = \sin x$ onto $y = 2\sin 3x$

j $y = \sin x$ onto $y = \sin(x - \frac{\pi}{3}) + 2$

c) Exercises : Modelling with sine functions

When patterns of variation can be identified and quantified using a formula or equation, predictions may be made about behaviour in the future. Examples of this include tidal movement which can be predicted many months ahead, and the date of a future full moon.

The graph over a two year period is shown below:



We attempt to model this data using the general sine function $y = a\sin(b(x-c)) + d$,
or in this case $T = a\sin(b(t-c)) + d$.

The period is 12 months, so $\frac{2\pi}{b} = 12$ and $\therefore \quad b = \frac{\pi}{6}$.

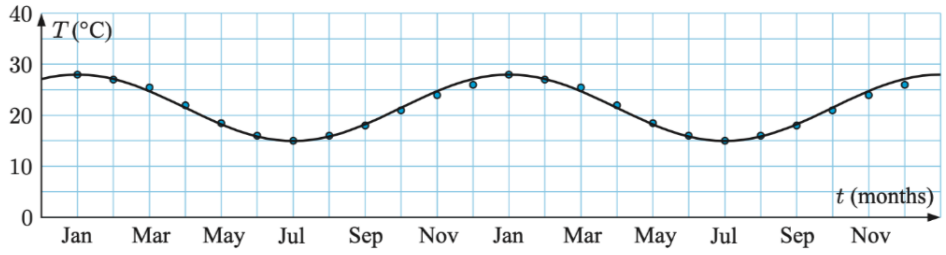
The amplitude $= \frac{\max - \min}{2} \approx \frac{28-15}{2} \approx 6.5$, so $a \approx 6.5$.

The principal axis is midway between the maximum and minimum, so $d \approx \frac{28+15}{2} \approx 21.5$.

So, the model is $T \approx 6.5\sin(\frac{\pi}{6}(t-c)) + 21.5$ for some constant c.

On the original graph, point A lies on the principal axis, and is the first point shown at which we are starting a new period. Since A is at $(10, 21.5)$, $c = 10$.

The model is therefore $T \approx 6.5\sin(\frac{\pi}{6}(t-10)) + 21.5$, and we can superimpose it on the original data as follows.



EXERCISE 9

Below is a table which shows the mean monthly maximum temperatures for a city in Greece.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Temperature (°C)	15	14	15	18	21	25	27	26	24	20	18	16

a Use a sine function of the form $T \approx a\sin(b(t-c)) + d$ to model the data. Find good estimates of the constants a, b, c and d without using technology. Use Jan $\equiv 1$, Feb $\equiv 2$, and so on.

b Use technology to check your answer to **a**. How well does your model fit?

EXERCISE 10

The data in the table shows the mean monthly temperatures for Christchurch, New Zealand.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Temperature (°C)	15	16	$14\frac{1}{2}$	12	10	$7\frac{1}{2}$	7	$7\frac{1}{2}$	$8\frac{1}{2}$	$10\frac{1}{2}$	$12\frac{1}{2}$	14

a Find a sine model for this data in the form $T \approx a\sin(b(t-c)) + d$. Assume Jan $\equiv 1$, Feb $\equiv 2$, and so on. Do not use technology.

b Use technology to check your answer to **a**.

EXERCISE 11

Some of the largest tides in the world are observed in Canada's Bay of Fundy. The difference between high and low tides is 14 metres, and the average time difference between high tides is about 12.4 hours.

a Find a sine model for the height of the tide H in terms of the time t.

b Sketch the graph of the model over one period.

III- Cosine functions

The cosine curve $y = \cos x$, like the sine curve $y = \sin x$, has a period of 2π, an amplitude of 1, and its range is $-1 \leq y \leq 1$.

Property

$\cos x = \sin\left(x + \frac{\pi}{2}\right)$.

Property

The general cosine function is

$$y = a\cos\big(b(x-c)\big) + d \qquad \text{where } a \neq 0,\ b > 0.$$

Example

On the same set of axes graph $y = \cos x$ and $y = \cos\left(x - \frac{\pi}{3}\right)$ for $-2\pi \leq x \leq 2\pi$.

Example

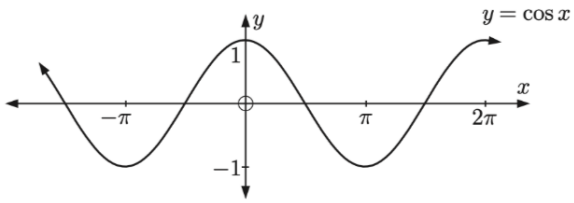
Without using technology, sketch the graph of $y = 3\cos 2x$ for $0 \leq x \leq 2\pi$.

Property

Since the cosine function is a horizontal translation of the sine function, the constants a, b, c, and d have the same effects as for the general sine function.

EXERCISE 12

Given the graph of $y = \cos x$, sketch the graphs of:



a $y = \cos x + 2$ **b** $y = \cos x - 1$ **c** $y = \cos\left(x - \frac{\pi}{4}\right)$

d $y = \cos\left(x + \frac{\pi}{6}\right)$ **e** $y = \frac{2}{3}\cos x$ **f** $y = \frac{3}{2}\cos x$

g $y = -\cos x$ **h** $y = \cos\left(x - \frac{\pi}{6}\right) + 1$ **i** $y = \cos\left(x + \frac{\pi}{4}\right) - 1$

j $y = \cos 2x$ **k** $y = \cos\left(\frac{x}{2}\right)$ **l** $y = 3\cos 2x$

EXERCISE 13

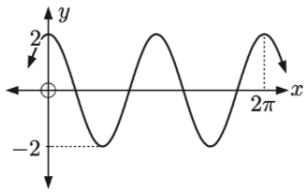
Without graphing them, state the periods of:

a $y = \cos 3x$ **b** $y = \cos\left(\frac{x}{3}\right)$ **c** $y = \cos\left(\frac{\pi}{50}x\right)$

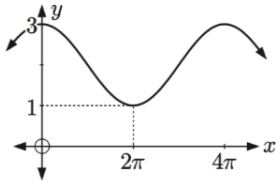
EXERCISE 14

Find the cosine function shown in the graph:

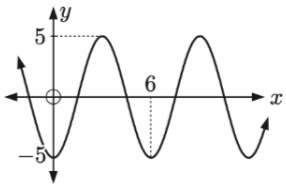
a



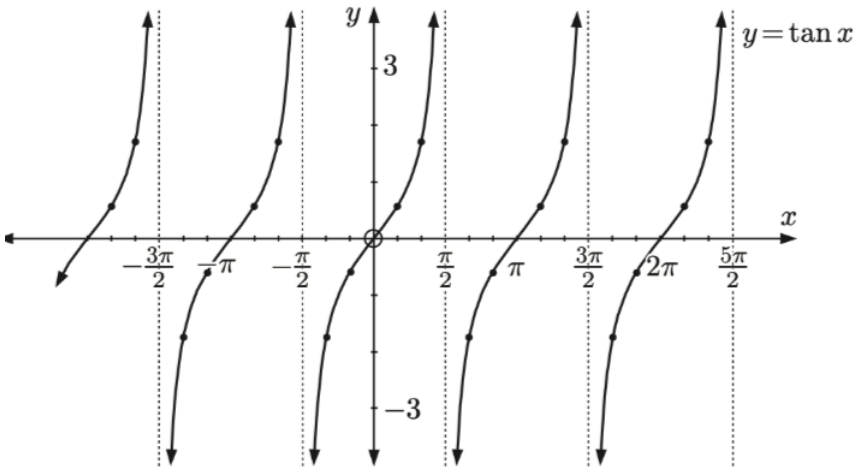
b



c



IV- Tangent functions



$y = \tan x$ has :

— period of π

— range $y \in \mathbb{R}$

— vertical asymptotes $x = \dfrac{\pi}{2} + k\pi$ for all $k \in \mathbb{Z}$.

Property

The general tangent function is $y = a\tan\big(b(x-c)\big) + d$, a$\neq 0,\ b > 0$.

— The principal axis is $y = d$.

— The period of this function is $\dfrac{\pi}{b}$.

— The amplitude of this function is undefined.

Example

Without using technology, sketch the graph of $y = \tan\left(x + \frac{\pi}{4}\right)$ for $0 \leq x \leq 3\pi$.

Example

Without using technology, sketch the graph of $y = \tan 2x$ for $-\pi \leq x \leq \pi$.

EXERCISE 15

a *Sketch* the following functions for $0 \leqslant x \leqslant 3\pi$:

i $y = \tan(x - \frac{\pi}{2})$ **ii** $y = -\tan x$ **iii** $y = \tan 3x$

b Use technology to check your answers to **a**.
Look in particular for asymptotes and the x-intercepts.

EXERCISE 16

Describe the transformation(s) which moves the first curve to the second curve:

a $y = \tan x$ to $y = \tan(x - 1) + 2$ **b** $y = \tan x$ to $y = -\tan x$

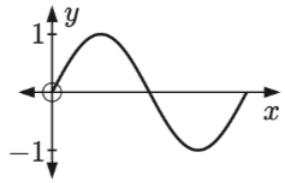
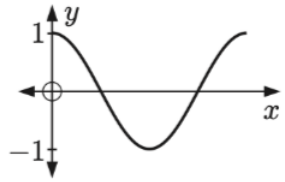
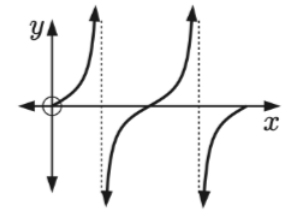
c $y = \tan x$ to $y = 2\tan\left(\frac{x}{2}\right)$

EXERCISE 17

State the period of:

a $y = \tan x$ **b** $y = \tan 3x$ **c** $y = \tan nx, \quad n \neq 0$

V- General trigonometric functions

FEATURES OF CIRCULAR FUNCTIONS					
Function	*Sketch for* $0 \leqslant x \leqslant 2\pi$	*Period*	*Amplitude*	*Domain*	*Range*
$y = \sin x$		2π	1	$x \in \mathbb{R}$	$-1 \leqslant y \leqslant 1$
$y = \cos x$		2π	1	$x \in \mathbb{R}$	$-1 \leqslant y \leqslant 1$
$y = \tan x$		π	undefined	$x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots.$	$y \in \mathbb{R}$

GENERAL TRIGONOMETRIC FUNCTIONS				
General function	a affects vertical stretch	$b > 0$ affects horizontal stretch	c affects horizontal translation	d affects vertical translation
$y = a\sin(b(x-c)) + d$ $y = a\cos(b(x-c)) + d$	amplitude $= \lvert a \rvert$	period $= \frac{2\pi}{b}$	• $c > 0$ moves the graph right • $c < 0$ moves the graph left	• $d > 0$ moves the graph up • $d < 0$ moves the graph down • principal axis is $y = d$
$y = a\tan(b(x-c)) + d$	amplitude undefined	period $= \frac{\pi}{b}$		

EXERCISE 18

State the amplitude, where appropriate, of:

a $y = \sin 4x$ **b** $y = 2\tan(\frac{x}{2})$ **c** $y = -\cos(3(x - \frac{\pi}{4}))$

EXERCISE 19

State the period of:

a $y = -\tan x$ **b** $y = \cos(\frac{x}{3}) - 1$ **c** $y = \sin(2(x - \frac{\pi}{4}))$

EXERCISE 20

Find b given:

a $y = \sin bx$ has period 2π **b** $y = \cos bx$ has period $\frac{2\pi}{3}$

c $y = \tan bx$ has period $\frac{\pi}{2}$ **d** $y = \sin bx$ has period 4

EXERCISE 21

Sketch the graphs of these functions for $0 \leqslant x \leqslant 2\pi$:

a $y = \frac{2}{3}\cos x$ **b** $y = \sin x + 1$ **c** $y = \tan(x + \frac{\pi}{2})$

d $y = 3\cos 2x$ **e** $y = \sin(x + \frac{\pi}{4}) - 1$ **f** $y = \tan x - 2$

EXERCISE 22

State the transformation(s) which map(s):

a $y = \sin x$ onto $y = \frac{1}{2}\sin x$ **b** $y = \cos x$ onto $y = \cos(\frac{x}{4})$

c $y = \sin x$ onto $y = -\sin x$ **d** $y = \cos x$ onto $y = \cos x - 2$

e $y = \tan x$ onto $y = \tan(x + \frac{\pi}{4})$ **f** $y = \sin x$ onto $y = \sin(-x)$

EXERCISE 23

Find m and n given the following graph is of the function $y = m\sin x + n$.

